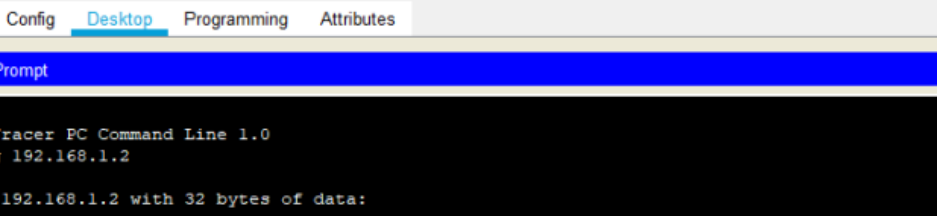


Practical 1a



The screenshot shows a Packet Tracer PC Command Prompt window for PC0. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt displays the following text:

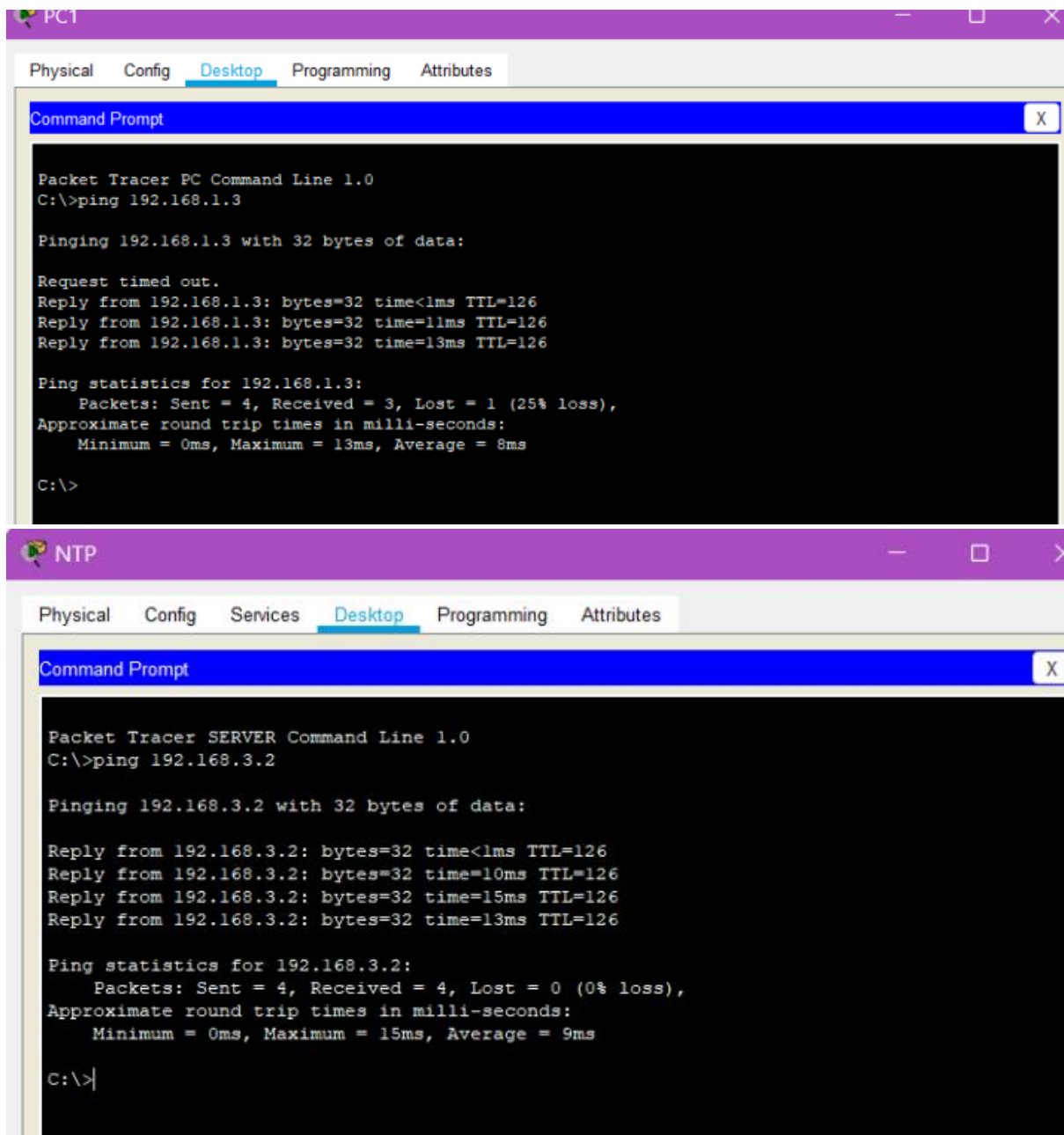
```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.3.1: Destination host unreachable.
Reply from 192.168.3.1: Destination host unreachable.
Reply from 192.168.3.1: Destination host unreachable.
Reply from 192.168.3.1: Destination host unreachable.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```



Practical 1b



```

Router>ena
Router>enable
Router#show clock
*0:5:7.120 UTC Mon Mar 1 1993
Router#

```

NTP

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP**
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

Service: NTP ☒ On ☐ Off

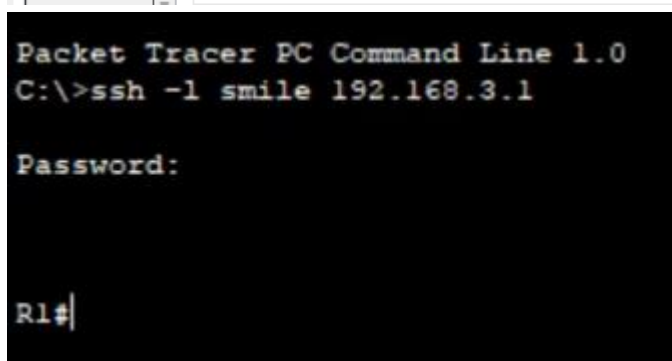
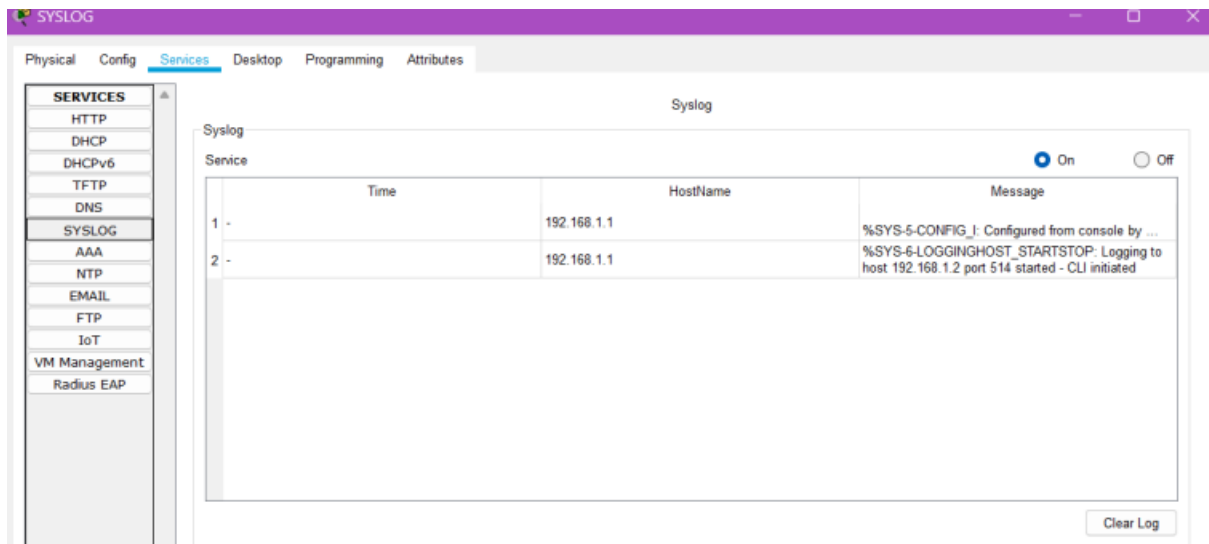
Authentication

☐ Enable ☒ Disable

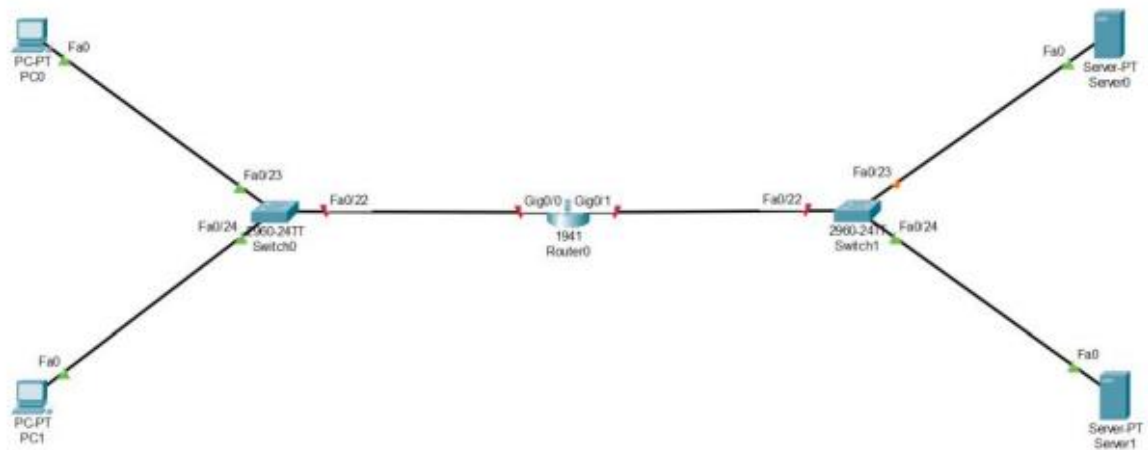
Key: Password:

March, 2023 03:50:18PM

Mon	Tue	Wed	Thu	Fri	Sat	Sun
27	28	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19



Practical 2



Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA**
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

AAA

Service ☒ On ☐ Off Radius Port 1645

Network Configuration

Client Name Client IP

Secret ServerType Radius

	Client Name	Client IP	Server Type	Key	
1	smile	192.168.2.1	Tacacs	abcd	Add Save Remove

User Setup

Username Password

	Username	Password	
1	smile	wxyz	Add Save Remove

Server1

Physical
Config
Services
Desktop
Programming
Attributes

SERVICES

HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP

AAA

Service
☒ On
☐ Off

Radius Port
1645

Network Configuration

Client Name
Client IP

Secret
ServerType
Radius

	Client Name	Client IP	Server Type	Key	
1	smile	192.168.2.1	Radius	abcd	Add Save Remove

User Setup

Username
Password

	Username	Password	
1	smile	wxyz	Add Save Remove

```

Packet Tracer PC Command Line 1.0
C:\>telnet 192.168.2.1
Trying 192.168.2.1 ...Open

```

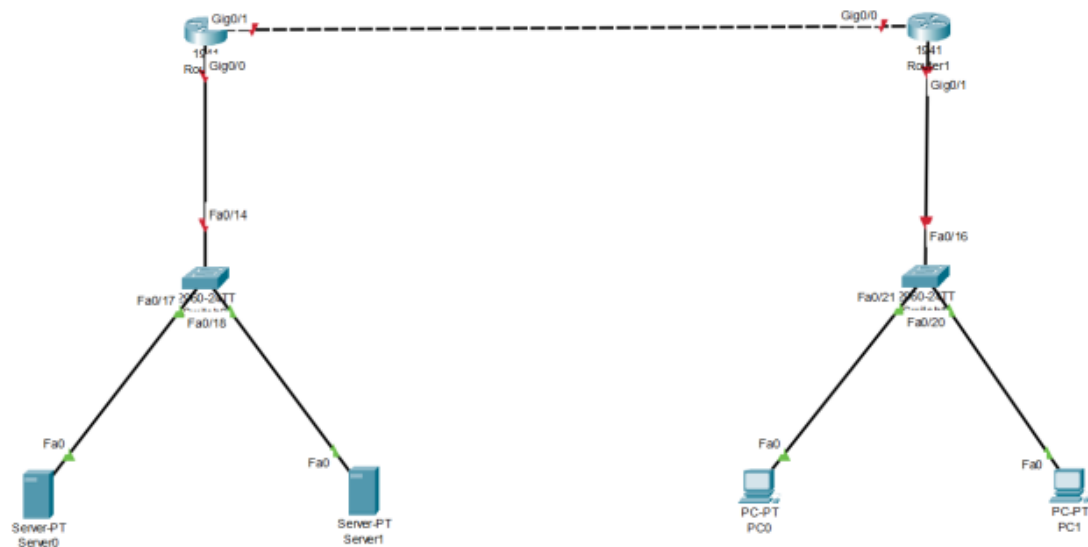
```
User Access Verification
```

```

Username: ismail
Password:
Router>

```

Practical3



```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=24ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 24ms, Average = 8ms

C:\>
```

```
Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time=3ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>
```

```

C:\>ftp 192.168.1.2
Trying to connect...192.168.1.2
Connected to 192.168.1.2
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>

```

```

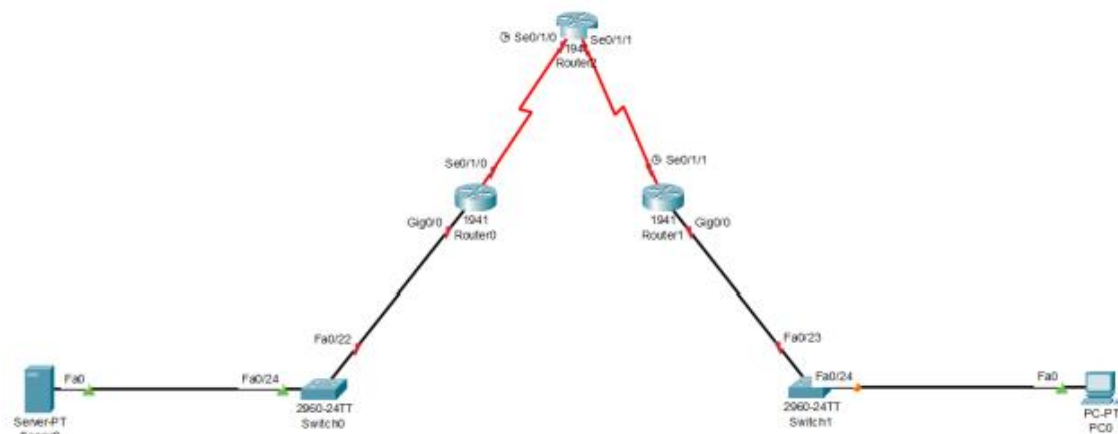
C:\>ftp 192.168.1.2
Trying to connect...192.168.1.2

%Error opening ftp://192.168.1.2/ (Timed out)
.

(Disconnecting from ftp server)

```

Practical4



PC0

Physical Config **Desktop** Programming Attributes

Command Prompt

```

Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=18ms TTL=125
Reply from 192.168.1.2: bytes=32 time=12ms TTL=125
Reply from 192.168.1.2: bytes=32 time=12ms TTL=125
Reply from 192.168.1.2: bytes=32 time=12ms TTL=125

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 18ms, Average = 13ms

```


The screenshot shows a Packet Tracer window titled 'Server0' with tabs for Physical, Config, Services, Desktop, Programming, and Attributes. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The Command Prompt shows the output of a 'ping 192.168.4.2' command, indicating a 25% loss. Below this, a message states '[Connection to 192.168.3.2 closed by foreign host]'. The prompt then shows an attempt to run 'ssh -l admin 192.168.3.2', which is refused by the remote host. The prompt ends with 'R1#'. A separate terminal window below shows the same 'ssh' command being entered, resulting in a 'Connection refused by remote host' error.

```
Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.4.2: bytes=32 time=12ms TTL=125
Reply from 192.168.4.2: bytes=32 time=12ms TTL=125
Reply from 192.168.4.2: bytes=32 time=12ms TTL=125

Ping statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 12ms, Average = 13ms

[Connection to 192.168.3.2 closed by foreign host]
C:\>ssh -l admin 192.168.3.2

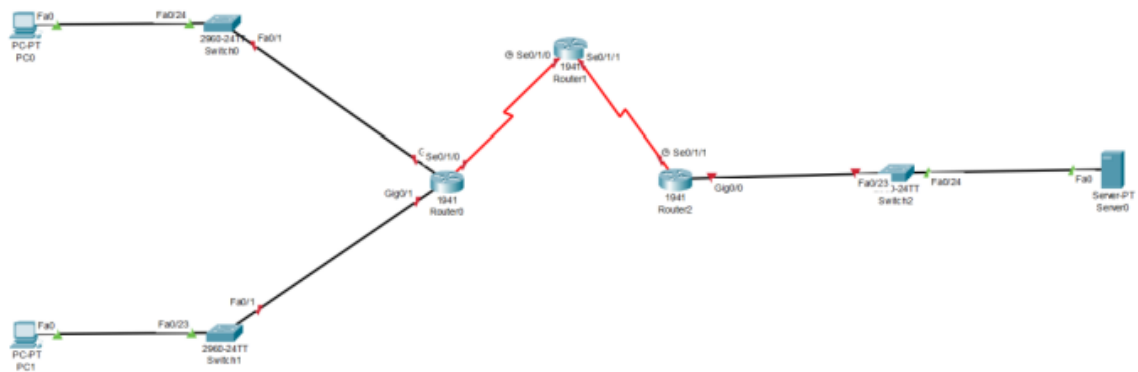
Password:

R1#

C:\>ssh -l admin 192.168.3.2

% Connection refused by remote host
C:\>
```

practical 5



```

Packet Tracer PC Command Line 1.0
C:\>ping 2005::2

Pinging 2005::2 with 32 bytes of data:

Reply from 2005::2: bytes=32 time=19ms TTL=125
Reply from 2005::2: bytes=32 time=5ms TTL=125
Reply from 2005::2: bytes=32 time=14ms TTL=125
Reply from 2005::2: bytes=32 time=11ms TTL=125

Ping statistics for 2005::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 19ms, Average = 12ms

C:\>

```

```

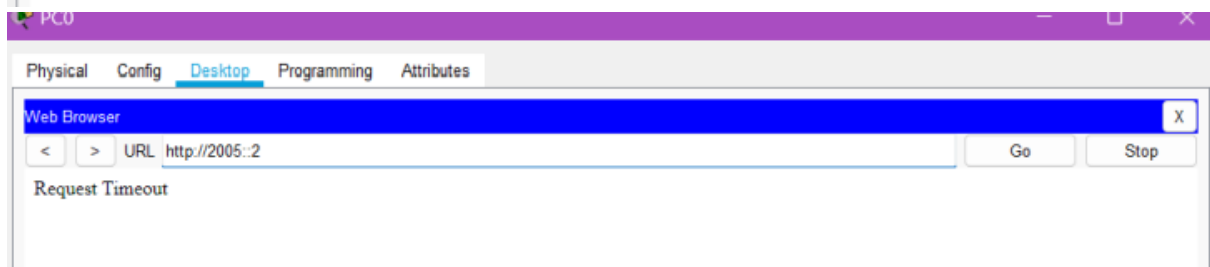
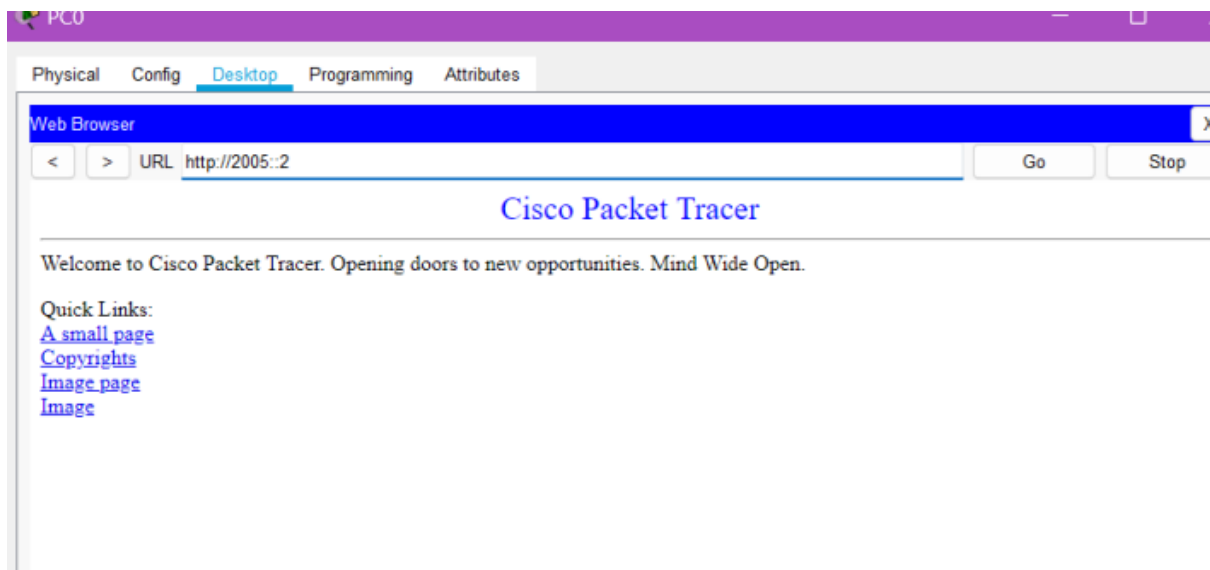
Packet Tracer SERVER Command Line 1.0
C:\>ping 2002::2/
Ping request could not find host 2002::2/. Please check the name and try again.
C:\>ping 2002::2

Pinging 2002::2 with 32 bytes of data:

Reply from 2002::2: bytes=32 time=3ms TTL=125
Reply from 2002::2: bytes=32 time=13ms TTL=125
Reply from 2002::2: bytes=32 time=13ms TTL=125
Reply from 2002::2: bytes=32 time=12ms TTL=125

Ping statistics for 2002::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 13ms, Average = 10ms

```



```
Packet Tracer PC Command Line 1.0
C:\>ping 2005::2

Pinging 2005::2 with 32 bytes of data:

Reply from 2005::2: bytes=32 time=3ms TTL=125
Reply from 2005::2: bytes=32 time=11ms TTL=125
Reply from 2005::2: bytes=32 time=17ms TTL=125
Reply from 2005::2: bytes=32 time=45ms TTL=125

Ping statistics for 2005::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 45ms, Average = 19ms

C:\>
```

Practical6



Router2

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/1/0

Serial0/1/1

Static Routes

Network 192.168.2.0

Mask 255.255.255.0

Next Hop 192.168.3.1

Add

Network Address

192.168.1.0/24 via 192.168.3.1

192.168.2.0/24 via 192.168.3.1

Router1

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/1/0

Serial0/1/1

Static Routes

Network 192.168.1.0

Mask 255.255.255.0

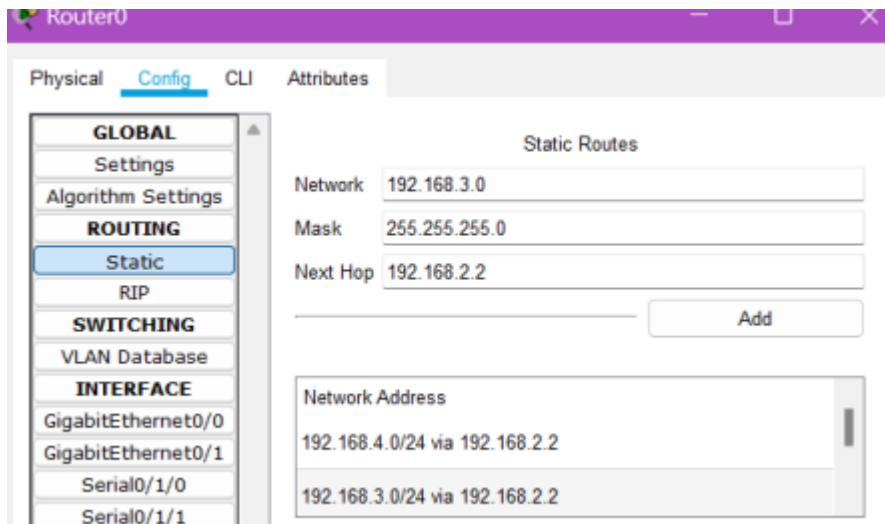
Next Hop 192.168.2.1

Add

Network Address

192.168.4.0/24 via 192.168.3.2

192.168.1.0/24 via 192.168.2.1



```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time=12ms TTL=125
Reply from 192.168.1.2: bytes=32 time=13ms TTL=125
Reply from 192.168.1.2: bytes=32 time=11ms TTL=125

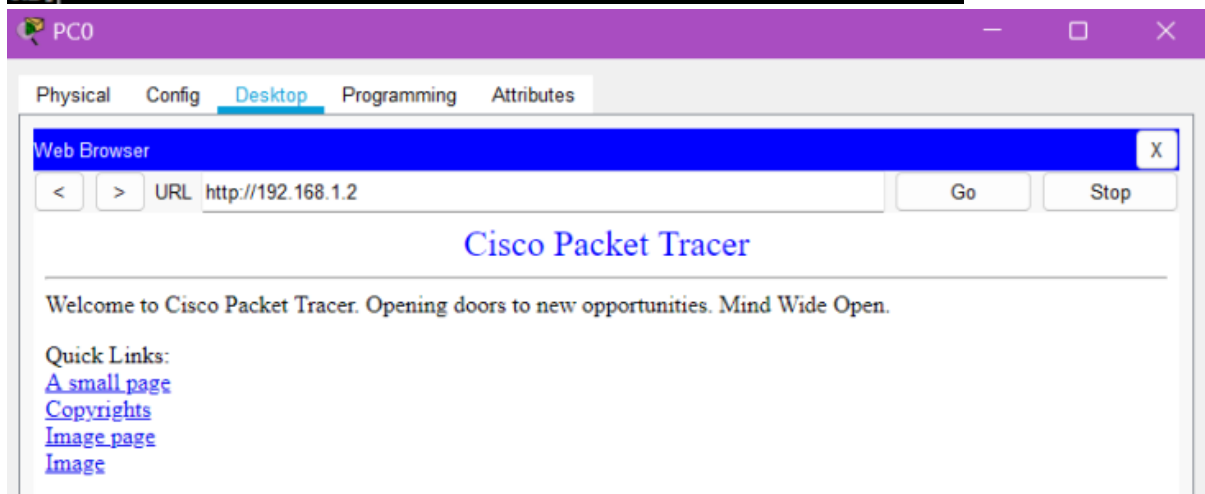
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 11ms, Maximum = 13ms, Average = 12ms

C:\>
```

```
C:\>
C:\>ssh -l ismile 192.168.3.1

Password:

R2#
```



Technology	Technology-package Current	Type	Technology-package Next reboot
ipbase	ipbasek9	Permanent	ipbasek9
security	None	None	None
data	None	None	None

Technology	Technology-package Current	Type	Technology-package Next reboot
ipbase	ipbasek9	Permanent	ipbasek9
security	securityk9	Evaluation	securityk9
data	disable	None	None

```
C:\>ping 192.168.1.2
```

```
Pinging 192.168.1.2 with 32 bytes of data:
```

```
Request timed out.
```

```
Reply from 192.168.1.2: bytes=32 time=41ms TTL=125
```

```
Reply from 192.168.1.2: bytes=32 time=13ms TTL=125
```

```
Reply from 192.168.1.2: bytes=32 time=11ms TTL=125
```

```
Ping statistics for 192.168.1.2:
```

```
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 11ms, Maximum = 41ms, Average = 21ms
```

```
C:\>ssh -l ismile 192.168.3.1
```

```
Password:
```

```
R2#
```

```
Class-map: in-map (match-all)
```

```
  Match: access-group 101
```

```
  Inspect
```

```
    Number of Established Sessions = 1
```

```
    Established Sessions
```

```
      Session 265448016 (192.168.4.2:1029)=>(192.168.3.1:22)
```

```
> SIS_OPEN/TCP_ESTAB
```

```
    Created 00:03:50, Last heard 00:03:47
```

```
    Bytes sent (initiator:responder) [741:725]
```

```
Class-map: class-default (match-any)
```

```
  Match: any
```

```
  Drop (default action)
```

```
    0 packets, 0 bytes
```

Web Browser

X

<

>

URL

http://192.168.1.2

Go

Stop

Cisco Packet Tracer

Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

Quick Links:

[A small page](#)

[Copyrights](#)

[Image page](#)

[Image](#)

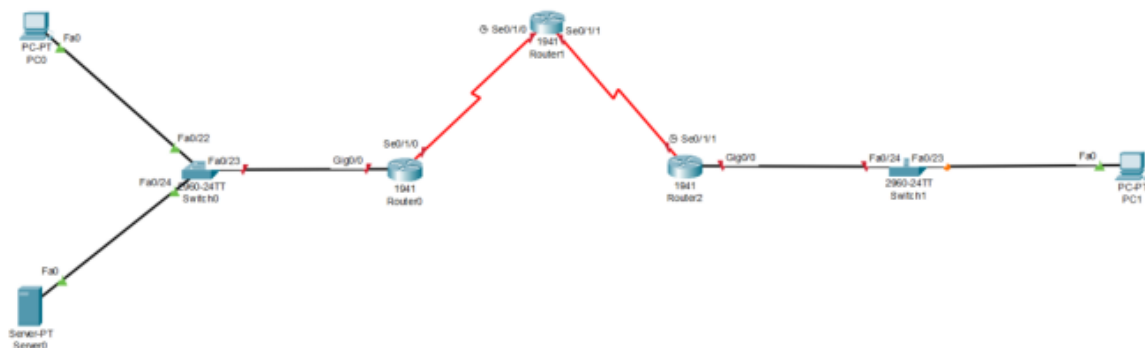
```
C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Practical 7



```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```

Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

Reply from 192.168.4.2: bytes=32 time=3ms TTL=125
Reply from 192.168.4.2: bytes=32 time=2ms TTL=125
Reply from 192.168.4.2: bytes=32 time=11ms TTL=125
Reply from 192.168.4.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 11ms, Average = 4ms

```

Technology	Technology-package Current	Type	Technology-package Next reboot
ipbase	ipbasek9	Permanent	ipbasek9
security	None	None	None
data	None	None	None

Technology	Technology-package Current	Type	Technology-package Next reboot
ipbase	ipbasek9	Permanent	ipbasek9
security	securityk9	Evaluation	securityk9
data	disable	None	None

```

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

```

C:\>ping 192.168.4.2

Pinging 192.168.4.2 with 32 bytes of data:

Reply from 192.168.4.2: bytes=32 time=3ms TTL=125
Reply from 192.168.4.2: bytes=32 time=11ms TTL=125
Reply from 192.168.4.2: bytes=32 time=2ms TTL=125
Reply from 192.168.4.2: bytes=32 time=12ms TTL=125

Ping statistics for 192.168.4.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 12ms, Average = 7ms

```



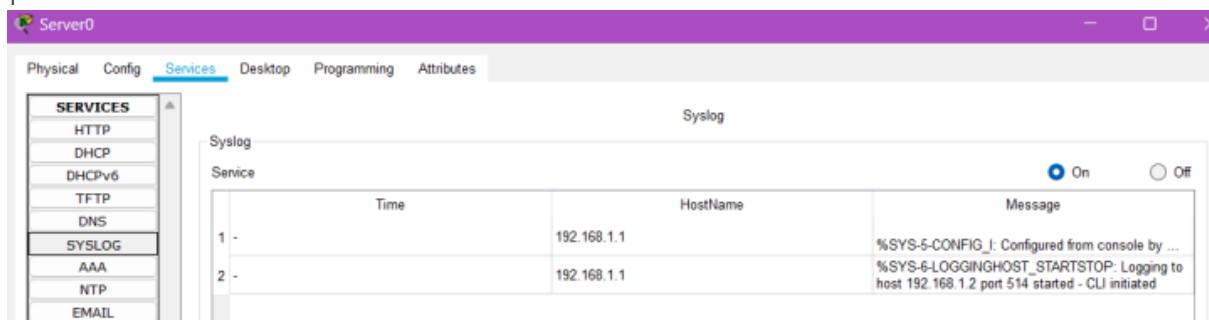
```

Router#conf
Router#configure ter
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#logging 192.168.1.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
%SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.2 port 514 started - CLI initiated

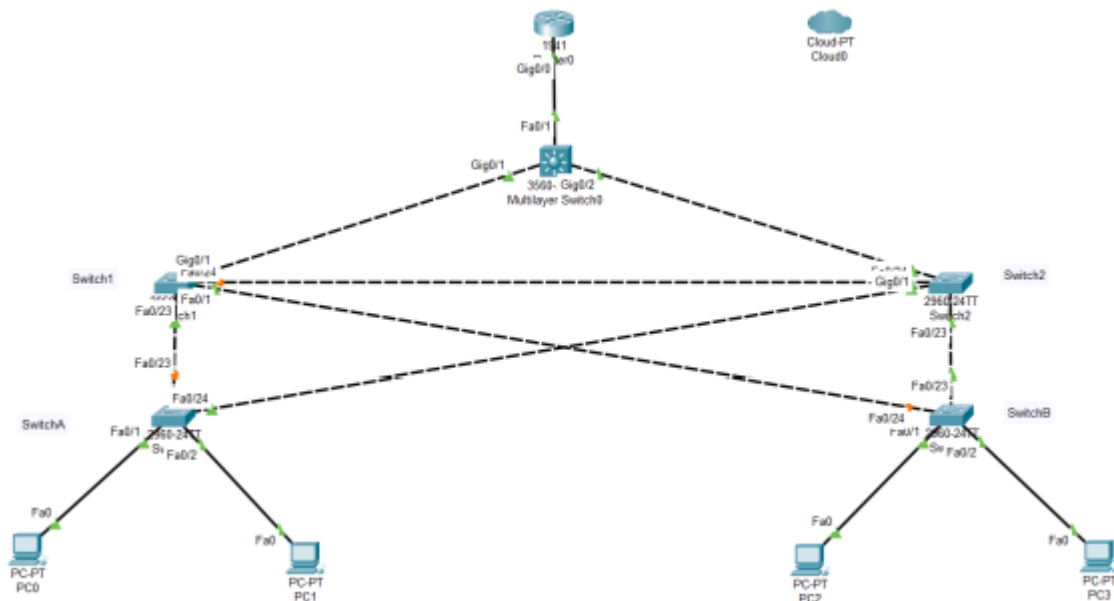
Router#ping 192.168.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/1/4 ms

```



Practical 8



Switch#show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769
 Address 000A.F389.8BE3
 Cost 23
 Port 26(GigabitEthernet0/2)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
 Address 0090.2B65.B15D
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/1	Altn	BLK	4	128.25	P2p
Gi0/2	Root	FWD	4	128.26	P2p
Fa0/1	Desg	FWD	19	128.1	P2p

Switch(config)#do show span

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 24577
 Address 0090.2B65.B15D
 This bridge is the root
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 24577 (priority 24576 sys-id-ext 1)
 Address 0090.2B65.B15D
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/1	Desg	FWD	4	128.25	P2p
Gi0/2	Desg	FWD	4	128.26	P2p
Fa0/1	Desg	FWD	19	128.1	P2p

```

VLAN0001
Spanning tree enabled protocol ieee
Root ID    Priority    24577
           Address    0090.2B65.B15D
           Cost        4
           Port        25(GigabitEthernet0/1)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    28673 (priority 28672 sys-id-ext 1)
           Address    0090.2BD1.1320
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Desg FWD	19	128.1	P2p	
Fa0/24	Desg LSN	19	128.24	P2p	
G10/1	Root FWD	4	128.25	P2p	
Fa0/23	Desg FWD	19	128.23	P2p	

```
SwitchA#show port-security interface f0/1
```

```

Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 2
Total MAC Addresses    : 0
Configured MAC Addresses : 0
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0

```

```
%LINK-5-CHANGED: Interface FastEthernet0/19, changed state to administratively down
```

```
%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to administratively down
```

```
%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down
```

```
%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to administratively down
```

```
SwitchA(config-if-range)#
```

```
%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down
```

```
%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to administratively down
```

```
%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down
```

```
%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down
```

```
%LINK-5-CHANGED: Interface FastEthernet0/18, changed state to administratively down
```

```
%LINK-5-CHANGED: Interface FastEthernet0/19, changed state to administratively down
```

```
%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to administratively down
```

```
%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down
```

```
%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to administratively down
```

```
SwitchB(config-if-range)#
```

```
%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down
```